

CS 766 COMPUTER VISION

Beating the Black Box: First-Layer Defect Detection

Catching print failures before they happen

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What is 3D Printing?

BACKGROUND

- Extrudes plastic layer by layer to build a physical physical part
- Creates low-cost 3D parts from a single roll of filament
- Used for prototypes, medical & dental parts, production runs, hobbyist models
- A single print takes hours — sometimes days — to complete



The Problem

3D printing is great! *But sometimes it's not...*

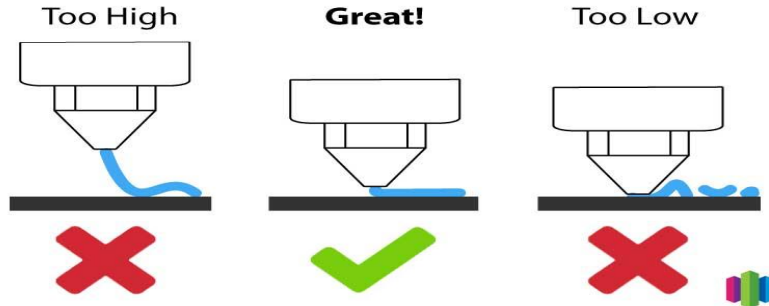
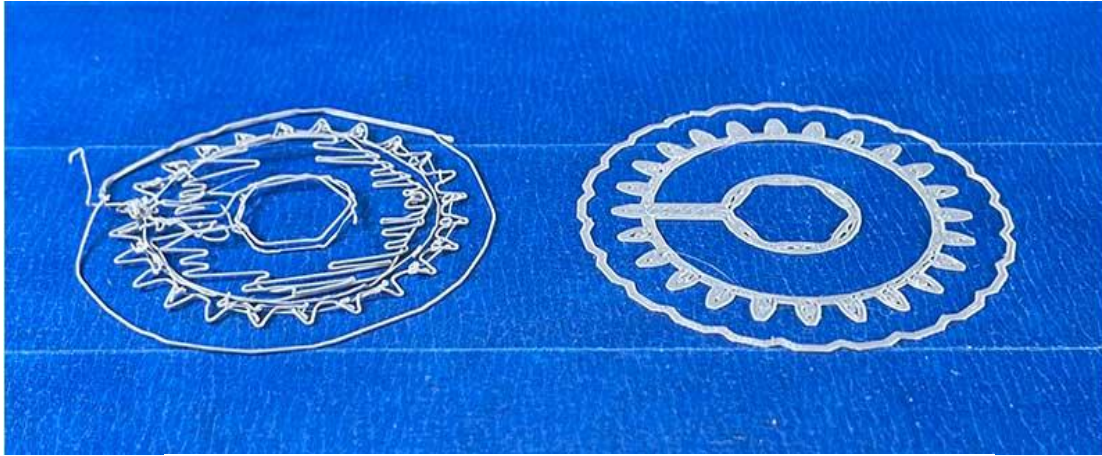


A 4-hour print, ruined by a bad layer

THE IMPACT

- Hours to days of lost time
- Wasted filament and resin
- Commercial settings: scrapped prototypes, delayed medical / dental parts, failed production runs
- A single failure can push a deadline or waste expensive material

Base Layer



- First step is the base layer
- This is the most difficult step
 - Plate misalignment
 - Plate prep (surface material, residue)
 - Plate temperature
 - Nozzle height
- A slight imperfection in base layer can lead to failures much later in print

The State of 3D Print Defect Detection

Common Defects

- Extrusion issues (under / over)
- Warping and layer shifts
- Adhesion loss
- Spaghetti / tangled filament
- Stringing and surface zits

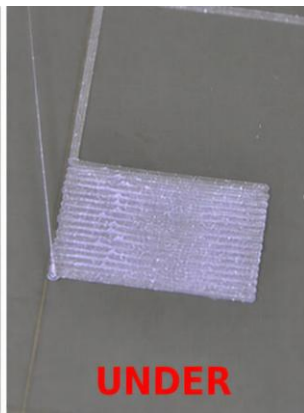
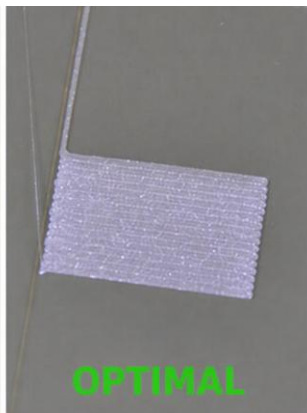
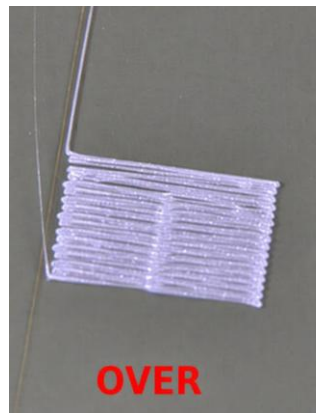
What Exists Today

- CNN tools like Obico / Spaghetti Detective
- Trained on catastrophic late-stage failures
- Reactive — fires once damage is done
- No proactive, early-stage detection

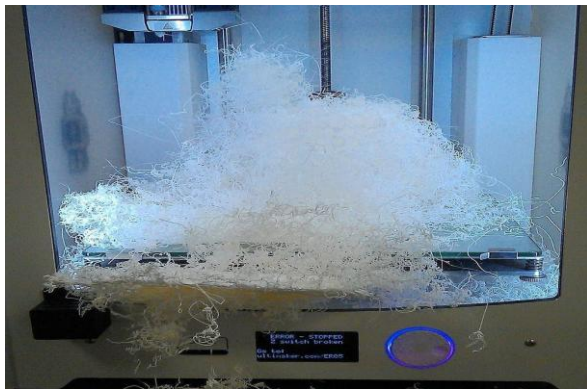
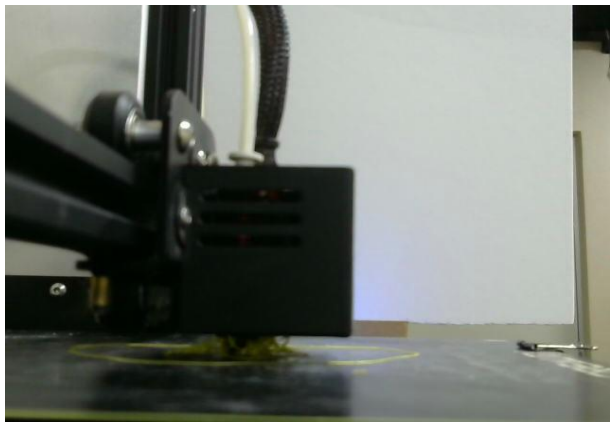
The gap: *nothing catches problems proactively, at the earliest possible stage.*

1st Layer categories

Trying to classify the base layer



Curating Model Dataset



Finding images was challenging

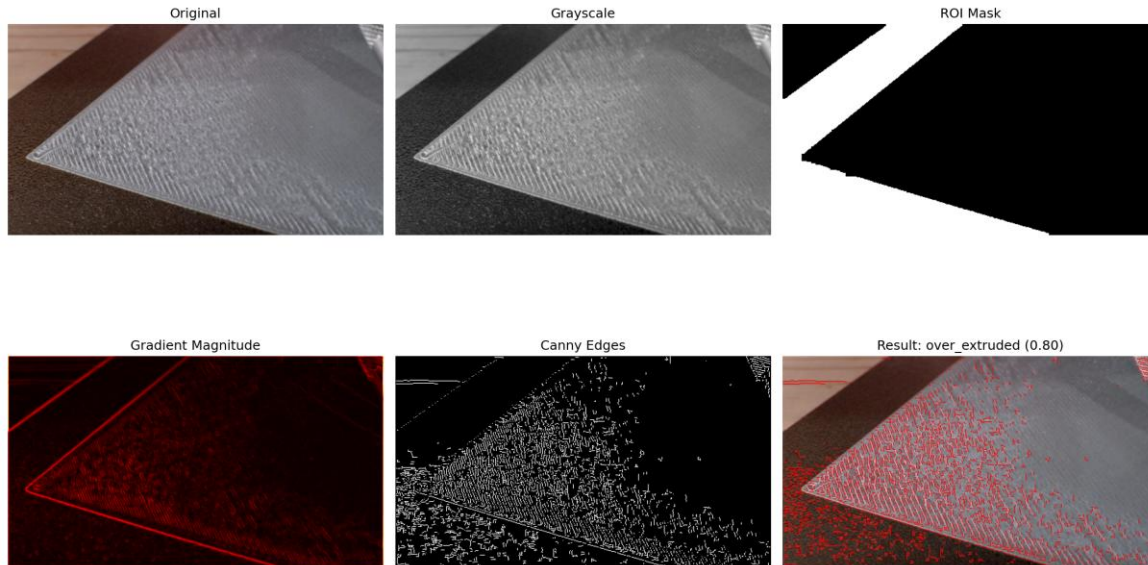
- Low Image quality
- Poor viewing angle
 - Top down is optimal
- Spaghetti prints
- Large unclassified image sets
- Instead we extracted images from Youtube videos of in progress prints and google images

Ground Truth Was Ours To Define



- We both classified every image independently
 - Over-extruded, under-extruded, optimal
- Disagreements → Moved to review folder or removed
- 16 images cut · 65 kept
- Manual labels ≠ perfect labels
- Needed to use our best judgement

Data Pipeline



6 Features → One Classification

All 6 features are combined into a single vector and fed into the classifier:

Fill density

% of ROI that is an edge pixel

Gradient entropy

spread of edge directions

Gradient kurtosis

peakedness of direction

Edge-to-gradient

edges vs. strong gradients

Edge density CV

variance across image patches

Line-spacing uniformity

regularity of extrusion lines

Line-spacing uniformity had largest iteration impact (+20% under-extruded recall)

Classifying images

How We Classified Each Image

1. Extract the 6-feature vector from the post-processed image
2. Compute the average feature vector μ_c for each class
3. Build a shared covariance matrix Σ to account for feature scale and correlation
4. Measure Mahalanobis distance from x to each class average
5. Assign the image to the class with the smallest distance

Classifier: Mahalanobis distance to each class centroid

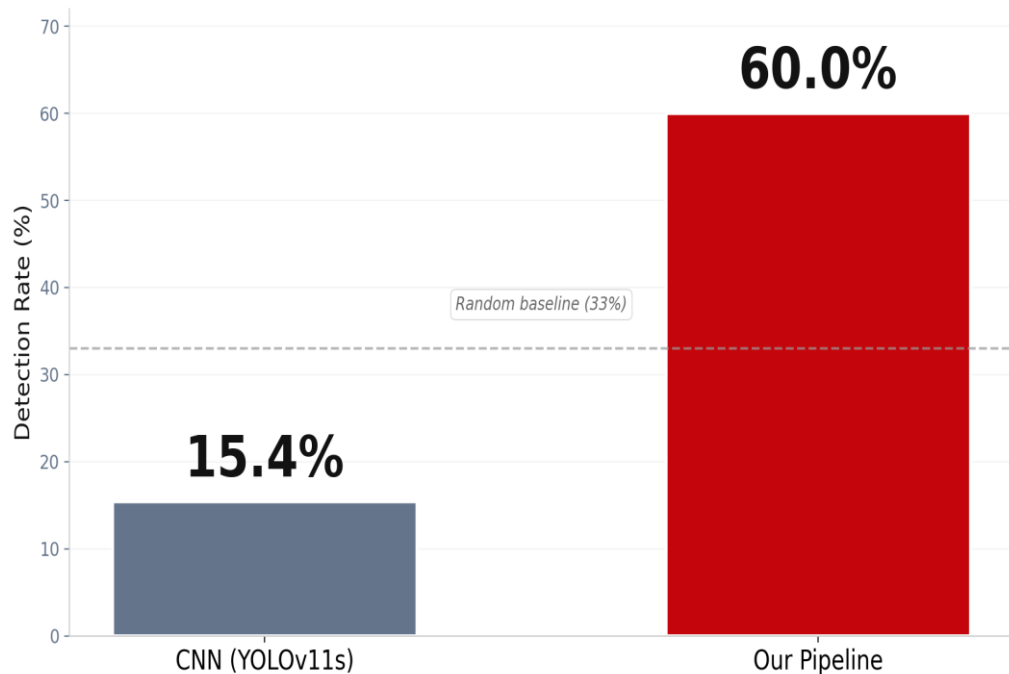
$$d(x, \mu_c) = \sqrt{(x - \mu_c)^T \Sigma^{-1} (x - \mu_c)}$$

x = 6-feature vector · μ_c = class mean · Σ = shared covariance (LDA)

pick class with smallest distance



Comparison to Other Models



YOLOv11s baseline

Trained on 9,000 images of spaghetti, stringing, and zits — **catastrophic failures** that physically cannot occur at layer 1.

The CNN isn't wrong —it's answering a question.

Our approach asks the right question at the right time.



Detection rate on 65 first-layer images · CNN misses 85% of defects at layer 1

What We Learned

Data Challenges

- **Top-down first-layer images** are rare — no off-the-shelf dataset existed
- Inconsistent lighting, camera angles, and build plates across sources
- We extracted most of our images manually from YouTube timelapse videos

Key Insight

- **Bottleneck is dataset size**
- With 65 images, ~8 features is near the classifier's limit
- Extra features overfit, even when physically meaningful

Future Work

- Collect a larger, more diverse labeled dataset
- **Integrate with printer firmware** so it auto-pauses when layer 1 looks bad
- Mobile app: point your phone at the printer and get an instant "keep going" or "stop"

Thank you.

Questions?

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github.com/jakeclifford127/cs766-3d-print-quality